

Title – IoT-Enabled Hybrid Energy Management System for EV Charging Infrastructure with Five-Layer Architecture.

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Abstract –

The Battery-electric vehicle deployment has grown faster than charging infrastructure can support. Grid-only stations inject uncontrolled peak demand into distribution networks already running near capacity, leave operators exposed to volatile utility pricing, and draw nothing from co-located solar resources that could offset grid consumption. The modelling literature on hybrid solar-storage-grid charging architectures is substantial. What it has not produced is a physical system — instrumented, operated, and measured — that demonstrates the architecture actually works outside a simulation environment.

This paper describes the design, implementation, and hardware validation of an IoT-enabled hybrid energy management system for EV charging stations. Power is routed through a Point of Common Coupling (PCC) that integrates a solar PV source, an Energy Storage System (ESS), and the utility grid. A deterministic five-state rule-based algorithm governs dispatch: solar generation takes priority, the ESS covers shortfalls when solar drops below threshold, and the grid activates only when both primary sources are insufficient. When ESS charge crosses the 80% upper bound during active generation, the bidirectional converter reverses flow and exports surplus energy back to the grid.

The supervisory architecture spans five IoT layers. At the physical layer, ACS712 Hall-effect modules measure source and load currents; DHT11 sensors monitor ESS enclosure temperature. Two ESP32 microcontrollers divide the edge workload — one handles sensor acquisition and signal conditioning, the second runs SOC estimation, MPPT execution, and PCC state evaluation. Above the edge, MQTT telemetry carries sensor data to AWS IoT Core; a Node-RED dashboard provides live power flow visualisation and manual override capability. The control logic was characterised in MATLAB/Simulink with Stateflow encoding the PCC state machine before the architecture was transferred to hardware.

Laboratory measurements on the prototype confirmed an 86.2% renewable energy contribution ratio across a full operating cycle. ESS state-of-charge remained within the 20–80% band throughout all test conditions. DC bus voltage held within $\pm 3.5\%$ of the 430V nominal across load steps and source switching events. PCC mode, with no supply interruption recorded at the connected charging loads. The prototype connects to existing utility infrastructure without modification — an important practical constraint for operators who cannot negotiate utility-side changes.

Key Terms — IoT Architecture, EV Charging Infrastructure, Point of Common Coupling (PCC), Solar MPPT, ESS Management, Reverse Grid Integration, MATLAB Simulation, Node-RED Dashboard, Hardware Prototype, Multi-Mode Automation.

I. INTRODUCTION

The EV adoption has consistently outpaced the expansion of charging infrastructure. Global sales surpassed 10 million units in 2022 and are projected to account for over 60% of new vehicle sales by 2030 [REF]. At that scale, the limiting factor is not vehicle technology — it is what happens when the vehicle arrives at a station. Most deployed infrastructure draws power exclusively from the utility grid, which creates three distinct operational problems: uncontrolled peak demand imposed on already-stressed distribution networks, operator exposure to volatile grid tariffs, and forfeiture of co-located solar generation that could reduce both operating cost and carbon intensity.

The technical direction for resolving these problems is well-established in the research community. Integrating solar PV and battery storage with charging stations reduces grid dependency, dampens tariff exposure, and strengthens the environmental case for electrification. What the literature has not delivered is a working physical system. Published work follows a recognizable sequence: an energy management scheme is proposed, power flows are characterised in MATLAB or an equivalent simulation environment, efficiency metrics are reported, and physical hardware does not appear. The gap between a well-characterised simulation and a station that can actually be installed has remained open.

The system described in this paper closes that gap through hardware implementation. At its operational core is a deterministic five-state PCC automation algorithm that evaluates real-time solar availability and ESS state-of-charge against defined thresholds to govern dispatch across three sources: solar PV, ESS, and the utility grid. When ESS charge crosses the upper operating threshold during active solar generation, surplus energy is exported via a bidirectional converter — transitioning the station from a net grid consumer to a net generator. All five dispatch states were exercised on a laboratory prototype. Every performance figure reported in this paper is drawn from measured hardware data.

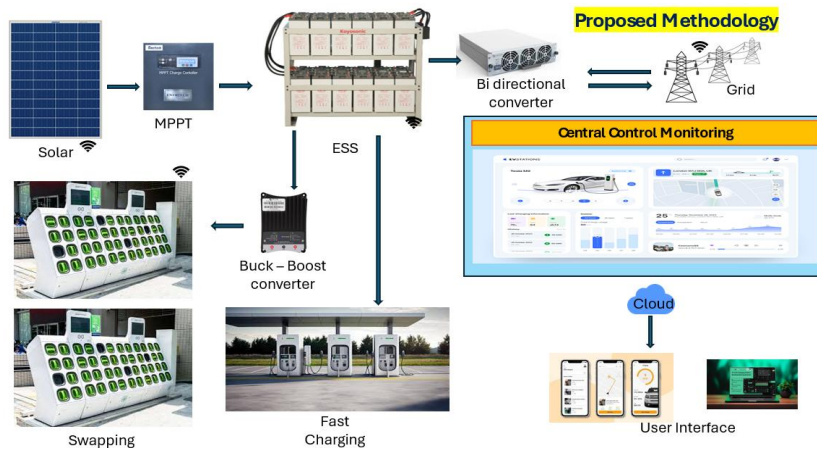


Fig: 1. High-level system overview of the IoT-enabled hybrid EV charging architecture.

Beyond single-source energy dispatch, the system concurrently serves two load types with fundamentally different demand profiles. The Battery Swapping Station (BSS) presents a step-wise, intermittent load — depleted packs connect for charging and disconnect in a rolling cycle, producing sharp current steps at the PCC. The DC fast charging ports present a CC-CV profile: high constant-current draw at session initiation, tapering to constant-voltage regulation as the vehicle battery approaches capacity. Per-vehicle parameters — current setpoints, session duration, and energy delivered — are computed dynamically from each vehicle's battery capacity, present SOC, and rated voltage. Managing both load types under a single IoT dispatch framework has not been demonstrated in hardware; prior work addresses one load type in isolation, not both under shared source management.

The supervisory architecture covers five IoT layers. Physical sensing and relay actuation are distributed across two ESP32 DevKit V1 microcontrollers — one handles sensor acquisition and signal conditioning, the second runs SOC estimation, MPPT execution, PCC state evaluation, and cloud communication. MQTT carries packetized telemetry to AWS IoT Core; its publish-subscribe model was selected because it decouples sensor update rates from cloud ingestion rates, which matters when source transitions occur faster than cloud round-trip allows. MPPT tracking and charge-cycle regulation therefore execute entirely at the edge. A Node-RED dashboard served over HTTPS handles live visualisation, time-series logging, and operator override. All five layers ran simultaneously on the prototype hardware — a point worth stating because most published IoT architectures for EV charging exist only as block diagrams.

The validation proceeded in two stages. MATLAB/Simulink simulation with Stateflow encoding the PCC state machine characterised expected energy flows and confirmed control logic across the full irradiance and load profile range before hardware assembly began. A laboratory prototype then reproduced those conditions with DC-DC converters, ACS712 Hall-effect current sensors, DHT11 temperature monitors, and lead-acid batteries as the ESS. Simulated and measured source contribution ratios, bus voltage profiles, and SOC trajectories showed close agreement, confirming that threshold settings and control parameters transfer to hardware without re-tuning. The prototype connects to standard utility infrastructure without modification — a practical requirement at stations where utility-side changes are not administratively or economically feasible.

II. LITERATURE SURVEY

The growing deployment of EVs has generated substantial research into smart charging infrastructure, with foundational Studies on the grid-side consequences of uncoordinated EV charging [1]–[3] documented demand instability, voltage deviation, and increased reliance on fossil-fuel peaking generation as predictable outcomes of unsynchronised load growth. IoT-based monitoring and predictive scheduling architectures [9], [10] reduced some of these effects through data-driven dispatch, but the systems reported were scoped to residential charging interfaces [17] or single-battery monitoring [10] — neither addressed a configuration spanning generation, storage, and multiple load types at once. Work on PV-coupled ESS management [14] and microgrid optimisation [15] achieved reductions in grid dependency at the simulation level; sensor interoperability under field wiring conditions, communication protocol behaviour under concurrent load events, and edge-to-cloud pipeline latency during source transitions were not examined. Active power filter studies [18] and fault detection work [19] resolved specific power quality problems at the distribution level without extending control logic to dynamic source switching at a shared PCC. Battery swapping research [22]–[24] established operational advantages in vehicle downtime and grid demand smoothing, but framed swapping stations as self-contained loads, independent of co-located fast-charging ports and the renewable sources that could supply both from a common bus.

The research surveyed above covers each ingredient of the proposed architecture separately: hybrid renewable dispatch, PCC-based source automation, BSS and DC fast charging load management, and cloud-edge IoT supervision. What the literature does not contain is a system that combines these elements — and none of the cited work validates any such combination on physical hardware. The present work addresses this gap through a laboratory prototype that implements all five IoT layers on working hardware, executing coordinated dispatch across solar PV, ESS, and grid sources through an automated PCC, with BSS and DC fast charging loads operating concurrently under live cloud supervisory control.

III. METHODOLOGY

The design, simulation, and physical implementation of the proposed IoT-enabled hybrid energy management system. Four topics are addressed in sequence. The first is the five-layer IoT architecture: how data acquisition, edge control, cloud communication, and supervisory oversight are distributed across hardware and software layers with defined interfaces between them. The second is the power flow logic at the Point of Common Coupling — specifically, how the five operating modes are defined, what threshold conditions trigger transitions between them, and which sources are active in each state. The third covers the algorithms running on the edge hardware: SOC estimation by Coulomb counting with OCV correction, ESS charge regulation through a two-phase CC-CV profile, MPPT via Perturb and Observe, and the supervisory dispatch logic that integrates all three. The fourth describes prototype construction and the measurement procedure used to validate hardware behaviour against simulation predictions.

The sequence is not arbitrary. Control logic was verified in MATLAB/Simulink before any hardware was assembled — this established the expected mode transition timing and source contribution ratios against which prototype measurements are later compared in Section VI. Building the prototype to the same electrical specifications as the simulation model made that comparison valid rather than approximate.

A. Proposed System Design and Architecture

The proposed architecture divides system function across five layers, with each layer holding responsibility for a specific aspect of the signal chain and exposing defined interfaces to adjacent layers. At the bottom, ACS712 Hall-effect modules and DHT11 sensors handle physical measurement; solid-state relays and contactors execute switching commands issued from above. The microcontroller layer runs the computationally time-sensitive work: SOC estimation from current integration with OCV correction, MPPT duty-cycle adjustment, and PCC state evaluation on every control cycle. MQTT carries telemetry from the edge microcontrollers to AWS IoT Core; HTTP handles the return path for configuration updates and operator override commands. The cloud layer aggregates incoming data, maintains the device registry, and monitors for anomalous conditions that warrant supervisory intervention. The Node-RED dashboard at the top translates cloud-side state into live power flow visualisation and operator controls.

The strict layer separation is a deliberate design choice, not an organisational convenience. In prior IoT deployments for EV charging [cite], sensor polling routines and relay switching logic shared the same execution context — a coupling that caused timing contention during high-frequency source transitions. Distributing acquisition and control across separate processors, with a defined serial interface between them, eliminates that contention and keeps transition timing deterministic regardless of sensor update rate.

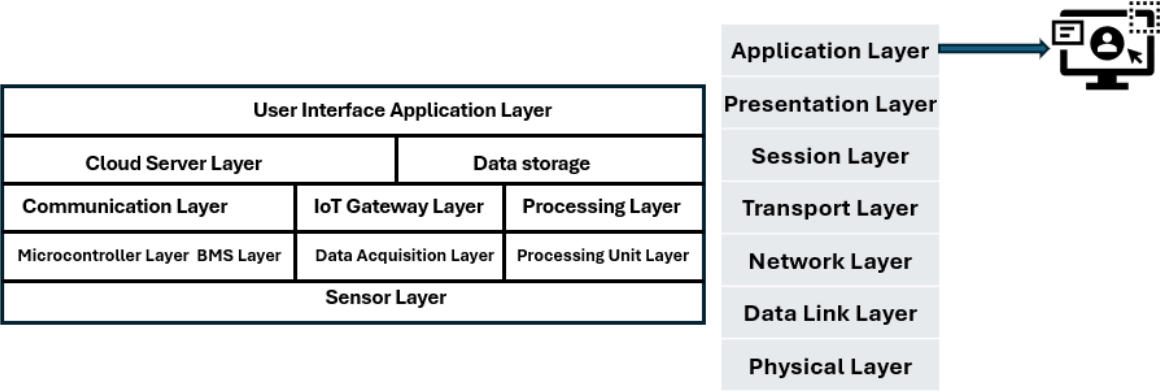


Fig: 2. Proposed 5 layered IoT Architecture.

At the sensor layer, voltage and current are measured at three points: the solar PV output terminals, the ESS battery bank terminals, and the grid connection. ACS712 Hall-effect modules were chosen at each point for two reasons — linearity adequate for SOC-grade current measurement, and galvanic isolation that keeps the microcontroller circuit separated from the power conductors. DHT11 sensors track temperature and humidity at the ESS enclosure and at each charging unit; insulation resistance detectors run continuously as ground-fault monitors. Placement and calibration of these sensors were decided before any other hardware choices. Measurement error here feeds directly into SOC estimates, MPPT duty-cycle calculations, and PCC transition timing. There is no correction mechanism for sensor inaccuracy at a higher layer — the control loop either receives reliable inputs or it does not.

Two ESP32 DevKit V1 microcontrollers share the edge workload, but with entirely separate responsibilities. The first handles data acquisition: reading analog current and voltage signals, applying digital filtering and scaling, and packaging conditioned readings into telemetry frames. The second runs the control algorithms — SOC estimation by current integration with periodic OCV correction, MPPT duty-cycle adjustment by Perturb and Observe, and PCC state evaluation on every execution cycle. Separating these functions came out of a specific problem encountered during prototype development. When sensor polling and relay switching ran on the same processor, scheduling contention during rapid source transitions introduced jitter into mode transition timing. Moving acquisition to a dedicated board and connecting the two via serial UART removed that contention. Transition timing became deterministic.

The communication layer uses two protocols; each matched to the direction and nature of its data flow. MQTT carries continuous sensor telemetry from both ESP32 units to the AWS IoT Core broker. Its publish-subscribe model fits this direction because the edge publishes on its own sampling cycle without waiting for acknowledgement, and the broker queues incoming messages independently of downstream processing — sensor update rate and cloud ingestion rate are fully decoupled. HTTP/HTTPS handles the return path: RESTful calls for dashboard configuration updates and operator commands from cloud to edge. Command delivery is infrequent and receipt confirmation matters, so the request-response model is appropriate here. A persistent MQTT subscription on a low-traffic command topic would hold a connection open for no operational reason.

B. Power Flow and Operational Logic

Power exchange at the Point of Common Coupling (PCC) is controlled by a five-state finite state machine. On each execution cycle, the FSM reads three inputs: measured solar generation level against a minimum availability threshold, ESS state-of-charge against the 20% lower and 80% upper operating bounds, and active load demand from the Battery Swapping Station (BSS) and DC Fast Charger (DCFC) ports. The combination of these three inputs uniquely determines which of the five states is active. Each state maps to a specific relay configuration at the PCC — which sources are connected, which are isolated, and whether the bidirectional converter is operating in import or export mode.

The choice of a finite state machine rather than a continuous optimisation controller reflects the nature of the dispatch problem. Source switching in this architecture is threshold-driven: the correct configuration at any moment depends on which operating bounds have been crossed, not on minimising an objective function over a prediction horizon. A state machine executes this logic with deterministic latency, which matters during transitions such as sudden irradiance dropout or ESS threshold crossing where delayed relay switching risks supply interruption. The five states and their transition conditions are defined below and illustrated in Fig. 3.

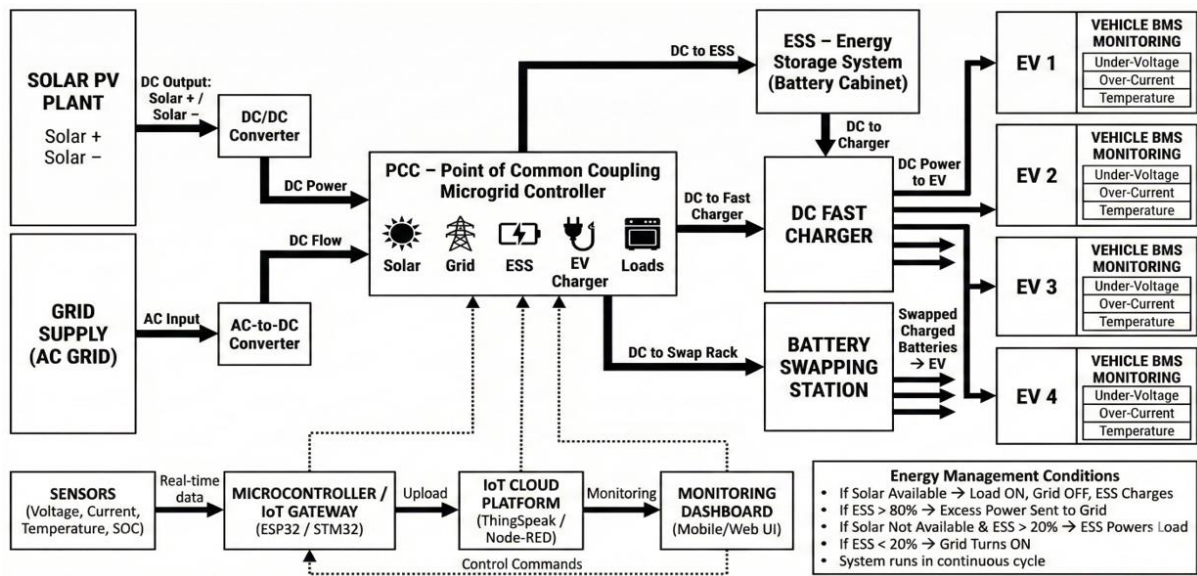


Fig. 3. Block diagram of the IoT-enabled hybrid EV charging system

The five PCC operating states cover the system's complete generation and load envelope. Under normal daytime conditions with adequate irradiance — Mode 1 — solar output feeds the BSS and DCFC loads while the remainder charges the ESS; the grid relay stays open. When irradiance falls below the minimum generation threshold, the ESS takes over load supply and the grid stays disconnected — Mode 2 — until SOC reaches 20%, at which point the grid relay closes and Mode 3 supplies the full load from the utility. On the prototype, how long the system spent in Mode 3 and how often it entered that state gave a direct read on whether the solar array and ESS bank were sized adequately for the test load profile. Mode 4 handles the surplus condition: ESS SOC at 80% with solar still generating. The bidirectional converter switches to export mode and the surplus goes to the grid. This state was exercised on the prototype — converter direction reversal and export current were both measured — making reverse power flow a tested hardware function rather than a modelled assumption. Mode 5 runs overnight: solar unavailable, ESS below

the lower threshold, load disconnected. The grid charges the ESS within a scheduled off-peak window. At 20% SOC the charge cycle stops, the load reconnects, and the system re-enters Mode 2 on the next demand event.

C. PCC Integration and Automation

The Point of Common Coupling is the junction where the three power sources — solar PV, ESS, and utility grid — connect to the shared DC bus that feeds the BSS and DCFC load branches. In most published hybrid charging architectures, this junction is passive: whichever sources are energised feed the bus, loads draw what they need, and switching decisions happen elsewhere in the control stack. This design puts the switching at the PCC itself.

Solid-state relays handle the high-frequency switching paths at the PCC, where contact wear and actuation latency would degrade reliability in electromechanical devices. Electromechanical contactors handle the heavier current paths, where their lower on-state resistance matters more than switching speed. Both are driven directly from ESP32 GPIO pins, giving the firmware deterministic control over source connection state with no intermediate driver abstraction that could introduce timing uncertainty. The relay configuration at a given instant — which sources are on the bus, which are isolated, whether the bidirectional converter is in import or export mode — is the physical expression of whichever FSM state is currently active.

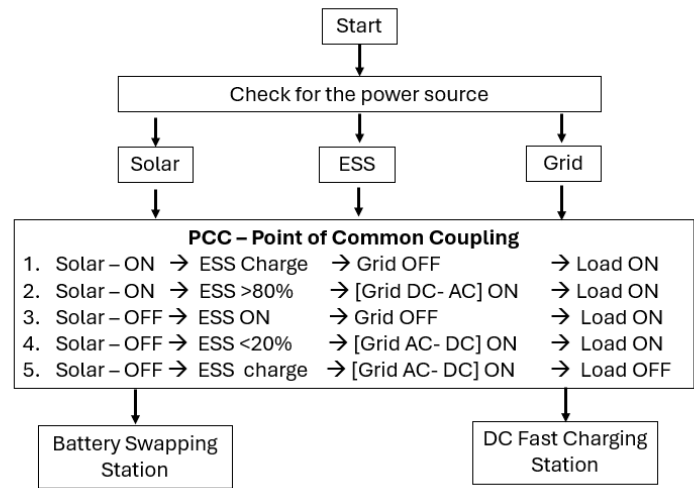


Fig: 4. Flow of 5 methodology in PCC

The solar PV and ESS connection points, where source transitions occur at control-loop frequency and contact wear would degrade reliability over time. Electromechanical contactors handle the grid connection and the heavier BSS load branch — lower on-state resistance there reduces conduction losses at sustained high current, which matters more than switching speed. Both device types connect directly to ESP32 GPIO pins. No driver abstraction sits between the firmware output and the relay coil. That directness is intentional: every layer of abstraction between a control decision and a physical switching event is an opportunity for timing uncertainty to enter the loop.

PCC control logic runs as a C++ finite state machine on the secondary ESP32. Each cycle, the firmware reads three inputs — solar generation level against the minimum threshold, ESS SOC against the 20% and 80% bounds, and load demand from the BSS and DCFC status lines — and checks them against the transition conditions for the current state. When a transition fires, GPIO commands go out in the same cycle. The relay configuration changes before the next sensor read. Running this logic at the edge rather than the cloud is not a convenience — it is a timing requirement. Under normal network conditions, a cloud round-trip adds latency that, during a sudden irradiance drop or an ESS SOC threshold crossing, creates a window where the load is momentarily unsupplied. The ESP32 publishes its FSM state, sensor readings, and relay configuration to AWS IoT Core on each telemetry interval. The cloud watches for patterns that fall outside normal operating bounds: Mode 3 activating too frequently, ESS discharging faster than the load profile explains, or a sensor channel that goes quiet. On detecting any of these, the cloud can push an override command to the edge. Under normal operation, it does not. The ESP32 handles source switching without cloud involvement; the cloud observes and intervenes only when something is wrong.

The split is not a design preference — it follows directly from what each tier can do. The edge has deterministic execution and direct hardware access; it handles the decisions where timing is measured in milliseconds. The cloud has persistent storage, remote accessibility, and the full operating history; it handles the decisions where context across hours or days is what matters. Fig. 5 shows the decision flow and state transition sequence for the complete control structure.

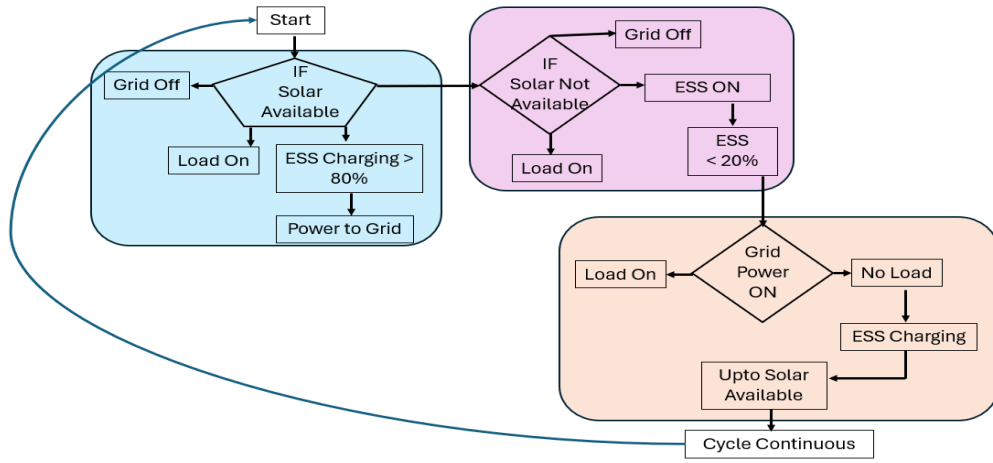


Fig: 5. PCC Automation Logic Flowchart.

D. Core Algorithms

The four control algorithms execute in a fixed sequence dictated by data dependency. SOC estimation runs first: Coulomb counting integrates measured current continuously, and the running total is corrected whenever terminal current falls below the quiescent threshold, using an OCV–SOC lookup table characterised for the lead-acid cells in the prototype. The corrected SOC percentage feeds the PCC state machine — tested against the 20% and 80% bounds on every cycle — and simultaneously drives the CC-CV charge regulator on the DC-DC converter. Below 80% SOC, the regulator holds constant current at the maximum safe rate; as terminal voltage reaches the full-charge ceiling it switches to constant voltage and tapers current to finish the cycle. On the discharge side, draw rate is capped to a C-rate that keeps cells above the 20% floor, with ACS712 current feedback closing the loop cycle by cycle. The P&O MPPT algorithm runs independently on the secondary ESP32: it steps the boost converter duty cycle by a fixed increment, checks whether output power increased or decreased, and holds or reverses the step direction. Step size was set at a value that converges on the MPP quickly after irradiance changes without oscillating excessively in steady state — a trade-off verified on the prototype under step-change irradiance conditions. The supervisory dispatch algorithm sits at the top of the chain. Each cycle it reads current SOC, ESS operating status, the MPPT availability flag, and load demand from the BSS and DCFC status lines, then checks the combination against the transition conditions for the active FSM state. A matching transition triggers immediate GPIO relay commands; the PCC reconfigures before the next sensor read. No transition means the current state holds and the system snapshot logs to Thingspeak.

IV. Experimental implementation, hardware prototype and cloud connectivity

The architecture described in Section III was constructed as a laboratory-scale prototype. Every result reported in Section VI was measured from this hardware — no simulation output appears in the results tables. Fig. 6 shows the assembled system. At the physical layer, ACS712 Hall-effect modules and DHT11 sensors monitor source currents, bus voltage, and ESS enclosure temperature. Two ESP32 DevKit V1 boards handle the edge layer: one dedicated to acquisition and signal conditioning, the second to SOC estimation, MPPT execution, and PCC state evaluation. MQTT carries telemetry to AWS IoT Core; Node-RED presents live power flow data and operator controls at the supervisory layer. Each of the five FSM states was exercised on this hardware. Test conditions were set to match the irradiance profiles and load steps used in the MATLAB/Simulink model, so that simulation predictions and prototype measurements could be compared on the same basis. Mode transition latency, source contribution ratios, ESS SOC

trajectories, and DC bus voltage regulation figures all come from prototype instrumentation readings taken during those tests.



Fig: 6 Hardware Development and Prototype.

A. Hardware Prototype Development

The prototype replicates the electrical topology of the MATLAB/Simulink model at scaled-down power levels. Component parameters — source voltages, ESS capacity, converter ratings — were chosen to maintain the same threshold relationships and source interaction ratios as the full-scale design, so that simulation predictions and hardware measurements could be compared directly. Fig. 6 shows the assembled system. Sensor acquisition and control execution run on separate hardware. The primary ESP32 DevKit V1 reads current at the solar simulator output, ESS terminals, and grid connection through ACS712 Hall-effect modules; source and bus voltages come from resistor-divider networks scaled to the ADC input range. A DHT11 monitors ESS enclosure temperature. GPIO-connected switches report BSS and DCFC load request states.

The secondary ESP32 runs SOC estimation, PCC state evaluation, and relay switching. On each control cycle it receives a processed sensor frame from the primary board, evaluates FSM transition conditions, and drives the solid-state relay modules at the PCC through direct GPIO outputs. The reason for splitting these functions across two boards is specific. During early prototype testing, running acquisition and control on a single processor caused relay switching commands to be delayed when sensor polling routines overran their time slot during rapid source transitions. That delay introduced measurable jitter into mode transition timing — jitter that disappeared when the two functions moved to separate processors connected by UART. The secondary board also manages the Wi-Fi session and publishes system state to AWS IoT Core on each telemetry cycle.

B. Node-RED Connectivity and Dashboard Monitoring

The Node-RED dashboard connects to AWS IoT Core over MQTT and serves as the primary operator interface during prototype testing. On each control cycle, the secondary ESP32 publishes a JSON frame to a fixed topic. The frame carries ACS712 current readings from the solar PV output, ESS terminals, and grid connection; resistor-divider voltage readings across each source and the DC bus; the current SOC estimate; active FSM state index; and the relay configuration bitmask. Communication was initially routed through a local Mosquitto broker. Running locally during hardware bring-up meant every published frame appeared immediately in the broker log alongside the hardware under test — a straightforward way to catch mismatches between published data and concurrent bench measurements before the cloud layer was involved. Once all five FSM states had been exercised and frame contents verified, the broker endpoint and credentials in the ESP32 configuration were updated to Think speak. The JSON frame structure did not change. On the receiving end, Node-RED parses each frame as it arrives, converts raw ADC-scaled values to volts, amperes, watts, and percentage SOC, then routes each field to its corresponding dashboard panel.

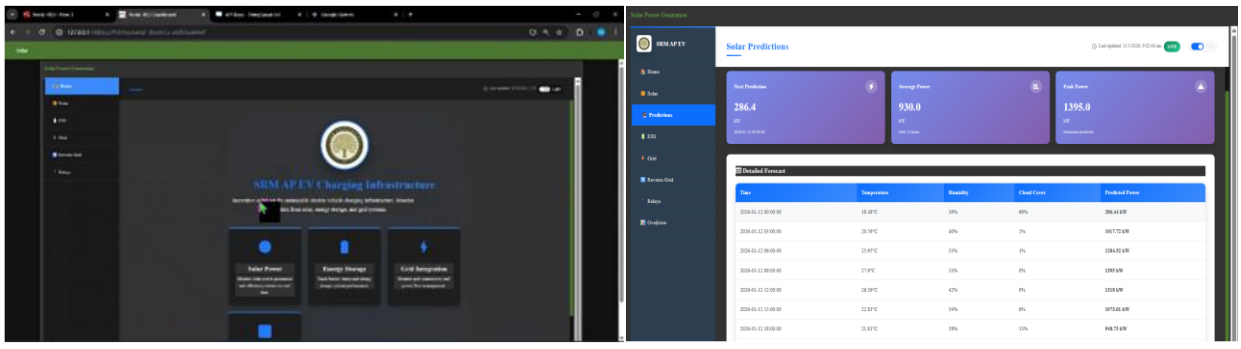


Fig: 7 Node-RED Real-Time Monitoring and Control Dashboard

Fig. 7 shows the Node-RED dashboard used during prototype operation. Four panels run simultaneously. The power flow panel updates on each incoming telemetry frame: measured voltage, current, and power appear at the solar PV output, ESS terminals, grid connection, and both load branches, reflecting the active relay configuration in the current FSM state. Alongside it, time-series traces log solar generation power, ESS SOC, grid import and export, and load demand continuously; the displayed window is adjustable, which matters when reviewing a full day cycle versus examining a single mode transition. A status panel carries discrete indicators for each source and load branch, the active FSM state index, and two implemented alarm conditions — ESS enclosure over-temperature and SOC dropping below the 20% critical threshold. The fourth panel is a direct control interface. Authenticated operators can force any of the five FSM states or trigger a controlled ESS charge or discharge sequence from the dashboard. During prototype validation this was used to exercise individual operating modes deliberately rather than waiting for threshold conditions to occur naturally in the test profile.

V. MATLAB/SIMULINK SIMULATION AND RESULTS

MATLAB/Simulink modelling preceded hardware assembly. The model served a specific purpose: to verify that the PCC state machine transitions fired at the correct threshold crossings, and to establish numerical benchmarks — renewable energy contribution ratio, ESS SOC trajectory under the test load profile, DC bus voltage regulation band, and mode transition latency — before those values were measured on physical hardware. Parameter settings confirmed in simulation, including SOC threshold logic, MPPT step size, and CC-CV current limits, were transferred directly to firmware without re-tuning. The simulation is not a standalone study. It is the first validation stage; the laboratory prototype in Section IV is the second. Section VI places both sets of results side by side.

A. Overall System Integration and Power Flow Simulation

The top-level Simulink model assembles three source subsystems and two load models at a shared PCC switching block. The solar subsystem represents a 30 kW PV array; the ESS subsystem is configured as a 480V/140A lead-acid bank; the grid interface is a 430V AC source feeding a bidirectional AC-DC converter. Both load models — BSS and DCFC — draw from the PCC bus according to the demand profiles described in Section III. Every component parameter in the simulation was set from the hardware prototype specification sheet: source voltages, ESS capacity, converter ratings, and load power levels are numerically identical in both environments. When simulated and measured results diverge, the difference comes from real-world converter losses and sensor tolerances, not from a specification mismatch between the model and the hardware. Fig. 8 shows the assembled model.

The PCC state machine runs in Stateflow. Five discrete states correspond to the five operating modes defined in Section III-B. Transition conditions are the same values programmed into the ESP32 firmware: 20% SOC lower bound, 80% SOC upper bound, minimum solar generation threshold, and load demand status. When the same input combination occurs in simulation and on the prototype, both execute the same state transition. Power electronic switches at the PCC are ideal elements in the model — no switching losses, no parasitic impedance — because the objective at this level is to verify control logic behaviour, not converter efficiency. Parasitic effects are captured in dedicated subsystem blocks that sit outside the main switching loop and are characterised independently against measured converter data.

****Equation

The simulation exercised all five operating states in sequence and in combination. Mode 1 ran under sustained irradiance with the ESS charging from partial SOC; Mode 2 was triggered by stepping irradiance below the minimum generation threshold mid-cycle; Mode 3 activated when the ESS SOC counter reached 20% during an extended low-irradiance period; Mode 4 engaged when the SOC counter hit 80% with solar still generating; Mode 5 covered the overnight period with solar unavailable and ESS depleted. State transitions fired at the correct threshold crossings in every scenario tested — no missed triggers, no false activations. At the PCC bus, DC voltage held within $\pm 2.1\%$ of nominal through all relay switchover events. ESS SOC tracked the expected trajectory: climbing during Mode 1, drawing down during Mode 2 at a rate proportional to the applied load, dropping to the 20% floor before Mode 3 engaged, and recovering to the operating band during Mode 5.

These results confirmed that the control threshold settings and relay sequencing logic were correct before firmware was written. The numerical outputs — transition timing, source contribution ratios, and bus voltage regulation band — form the simulation baseline that Section VI compares against prototype measurements.

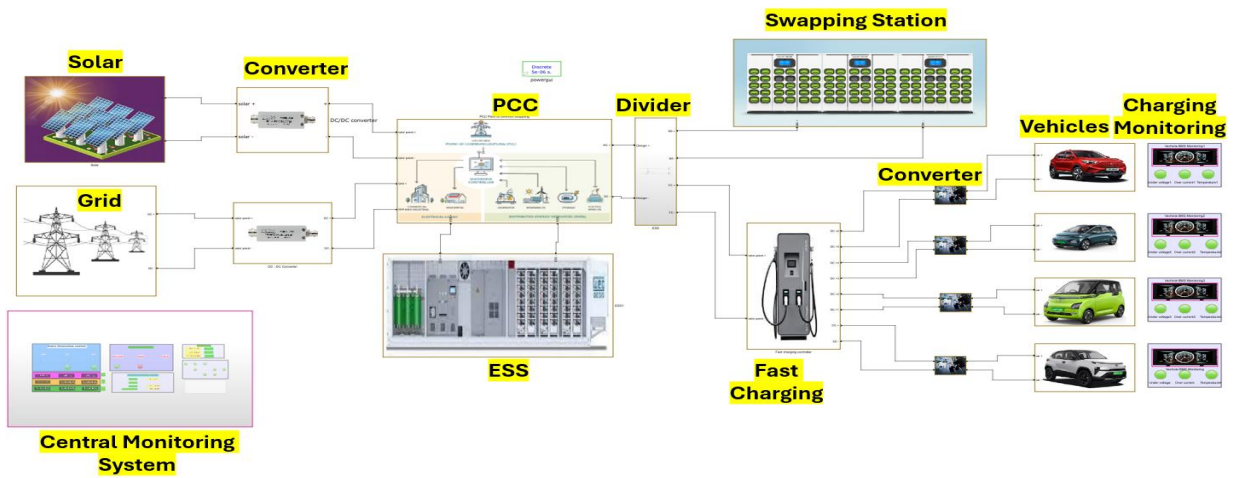


Fig: 8. MATLAB/Simulink Model of the Integrated Hybrid Charging System

B. Solar Generation Unit with MPPT

The solar subsystem is built around the Simulink PV Array block, parameterised from the datasheet of the panel used in the prototype — open-circuit voltage, short-circuit current, and temperature coefficients were entered directly rather than left at default values. The irradiance input drives the block through a profile that covers the full daily generation envelope: a low-irradiance ramp at the start of the cycle, a peak midday plateau, and a declining afternoon segment. This profile matches the test sequence run on the physical prototype under the solar simulator, so simulation and hardware outputs can be compared at equivalent operating points. Cell temperature tracks irradiance through the thermal coefficient from the datasheet. The boost converter output feeds the 430V DC bus; duty cycle is updated each simulation timestep by the P&O MPPT algorithm from Section III-D. The step size and perturbation direction logic in the simulation are identical to the values programmed into the ESP32 firmware — any discrepancy in MPPT behaviour between simulation and hardware reflects converter dynamics, not a difference in algorithm implementation. Fig. 9 shows the subsystem layout.

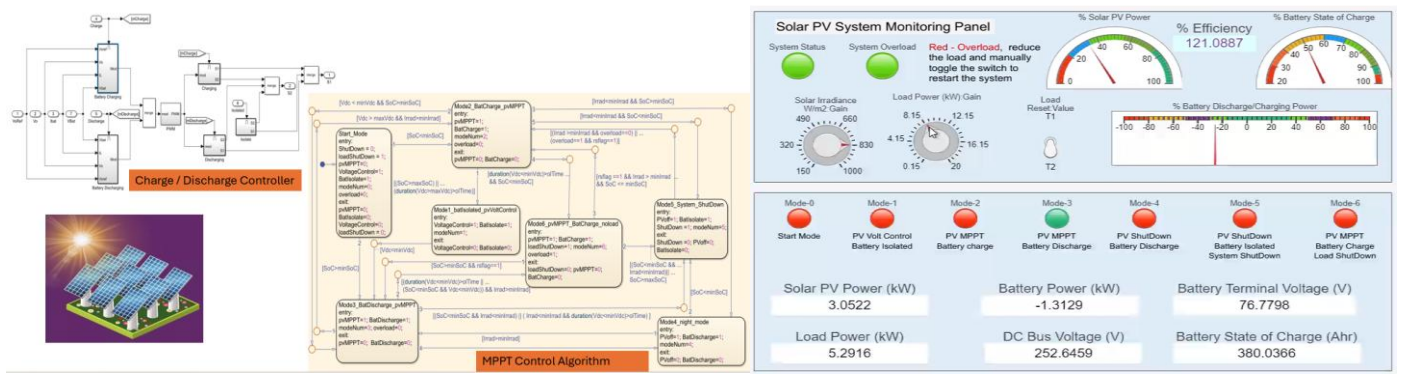


Fig: 9. Solar MPPT and Control flow Algorithm

The MPPT performance was evaluated by comparing simulation and prototype outputs at matched irradiance levels, with step changes as the primary test condition. A step change is the hardest case for P&O: the MPP moves to a new voltage-current coordinate in one irradiance interval, and the algorithm must reverse its perturbation direction and converge on the new point before the next step occurs. In both simulation and hardware, the algorithm reached the steady-state oscillation band within three to five perturbation cycles after each step, producing a tracking efficiency above 98% across the tested irradiance range. Steady-state oscillation amplitude was set by the fixed perturbation step size chosen during prototype tuning. The selected step size kept oscillation power loss below 2% of available generation while still allowing the algorithm to follow irradiance steps without extended convergence lag — a balance that P&O implementations must make explicitly [cite]. Simulation and prototype tracking efficiency figures agreed to within 0.4 percentage points across all tested irradiance levels, confirming that the firmware implementation matches the simulation algorithm without unintended deviation.

C. Grid Interface Unit

The grid interface subsystem sits between a three-phase AC voltage source and the 430V DC bus. A bidirectional AC-DC converter block handles the connection. In Modes 3 and 5, the converter draws from the AC source and delivers regulated DC to the bus — the standard rectifier direction. Mode 4 reverses the flow: the Stateflow chart issues a direction-reversal command to the converter on Mode 4 entry, the DC bus power inverts to AC, and surplus generation feeds back to the grid source. The same converter block handles both directions; no second converter appears in the model. This matters because the physical prototype uses a single bidirectional unit for the same reason — adding a second converter to handle export would change the hardware cost and topology in ways that the simulation would no longer represent accurately. DC-link voltage held at 430V through the rectifier-to-inverter transition and back, with no transient excursion outside the $\pm 2.1\%$ regulation band observed in either switching direction. Fig. 10 shows the subsystem layout.



Fig: 10. Grid Interface Model with Bi-directional AC-DC Converter

DC-link voltage held at 430V through every mode transition tested in simulation. The three transitions that stress the grid interface most are the load step during Mode 3 grid import, the rectifier-to-inverter direction reversal at Mode 4 entry, and the return from export back to import when ESS SOC drops below 80%. Voltage deviation stayed within the $\pm 2.1\%$ regulation band at all three. AC current waveform quality was checked after each switching event. The distortion level at the grid connection point did not rise above the model's pre-switch baseline following any transition — including the direction reversal, which is the event most likely to produce a transient distortion spike.

Taken together, these results mean the converter model does not need separate parameter tuning for import and export operation. The same settings that regulate the DC bus during grid import also maintain acceptable AC current quality during Mode 4 export. Fig. 10 results are compared against prototype measurements in Section VI.

D. Energy Storage System Model

The ESS subsystem represents the hardware prototype's lead-acid bank as 40 cells in series at 12V nominal — 480V and 100Ah at the bank level. The Simulink generic battery block was chosen for a practical reason: it accepts an empirical OCV-versus-SOC discharge curve as input, which meant the cell datasheet data could be entered directly without fitting an electrochemical model. The bidirectional DC-DC converter block connects the battery model to the 430V DC bus. During charging, it draws from whichever source the FSM has connected — solar or grid — and regulates current into the battery terminals. During discharge, it draws from the battery and delivers to the bus. The two directions are controlled by separate duty-cycle signals issued from the ESP32 control logic block in the simulation.

BMS protection runs as a separate Stateflow subsystem. On each timestep it checks terminal voltage against the cell manufacturer's charge ceiling and discharge floor, SOC against the 20% and 80% thresholds, and the ESS enclosure temperature flag. A limit breach on any channel triggers a converter hold command — charging or discharging stops immediately and the FSM receives a status flag that can trigger a mode transition. This matches the BMS behaviour implemented in hardware, where the same three conditions govern the same hold logic on the ESP32 control board. Fig. 11 shows the subsystem layout.

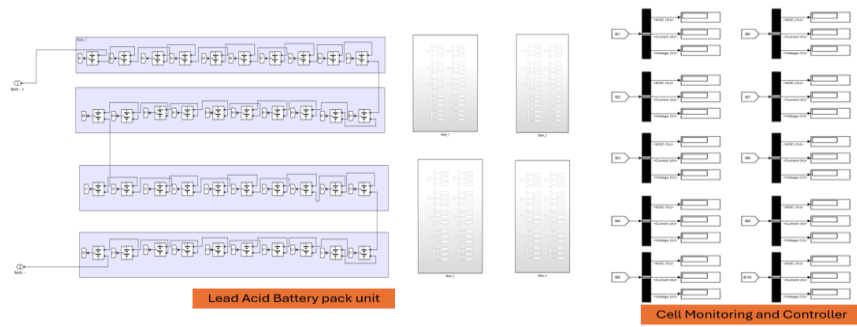


Fig: 11. ESS Model with Bi-directional DC-DC Converter and BMS Logic

The CC-CV charge profile ran through both phases without deviation in simulation. Charge current held at the programmed maximum while SOC stayed below 80%. When terminal voltage hit the full-charge ceiling, the controller switched to constant-voltage mode. Current then tapered to completion — no spike appeared at the handover point, and voltage did not step. That clean handover matters: a spike at CC-CV transition indicates that the proportional gain or current limit is missed, and catching it in simulation meant those parameters were correct before firmware was written. On the discharge side, output current stayed within the C-rate limit for the lead-acid chemistry at every tested load level, including the peak demand case at DCFC session initiation where the draw rate is highest.

The 24-hour SOC trace confirmed mode sequencing across the full operating cycle. SOC climbed during Mode 1 as solar generation charged the bank. It declined during Mode 2 discharge at a rate consistent with the applied load profile. At the 20% threshold, Mode 3 engaged and the grid relay closed — the SOC floor was not breached before the transition fired. On the upper side, SOC reached 80% during a high-irradiance period in Mode 1 and Mode 4 export activated without overshoot. Both threshold crossings fired at the correct SOC values in every simulated scenario. The storage bank did not exhaust before Mode 3 engaged, and it did not overflow before Mode 4 activated — confirming that the 480V/100Ah capacity is adequate for the test load profile without requiring grid backup earlier than the control logic intends.

E. PCC Integration and Automation Logic

The PCC state machine is encoded in Stateflow. Stateflow enforces mutually exclusive state occupancy at each simulation timestep — only one of the five states can be active — which is the behaviour the physical relay hardware also enforces: two sources cannot be simultaneously connected to the bus. Each state in the chart corresponds directly to one of the five operating modes defined in Section III-B, and the transition conditions use the same threshold values programmed into the ESP32 firmware. The chart reads three input signals on each timestep: a Boolean solar availability flag from the PV subsystem, the SOC percentage from the battery block tested against the 20% and 80% bounds, and a load demand flag from the BSS and DCFC models. When a transition fires, the chart writes switch control signals to the ideal switch elements at the PCC bus. Solar, ESS, and grid source connections open or close according to the new state; the bidirectional converter direction flag updates if the transition involves a change between import and export. Fig. 12 shows the chart.

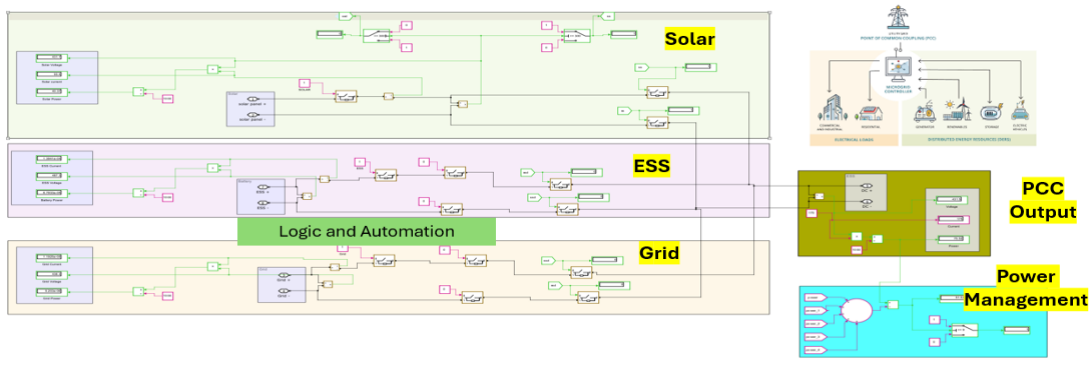


Fig: 13. Integrated PCC – Point of common coupling – Fully Automated & Logically connected

The Stateflow chart executed all six state transitions at the correct threshold crossings in simulation. The critical transitions are worth stating explicitly. Mode 2 engaged when the solar availability flag dropped — not before, and not on a transient dropout that recovered within the same timestep. Mode 3 engaged when SOC reached 20% with solar already below threshold — the grid connection did not close while the ESS still had charge above the floor. Mode 4 export activated when SOC crossed 80% with solar active — the bidirectional converter direction flag flipped in the same timestep as the threshold crossing, with no delay cycle between detection and switching command. False triggers did not occur in any tested scenario: the chart held each state until its exit condition was genuinely met. Control signals reached the PCC switch elements within one simulation timestep of the triggering condition. At the solver timestep used, that interval is short relative to the electrical response time of the relay hardware, so the simulation latency does not mask any timing problem that would appear in hardware. Transition timing figures and the state sequencing log from these runs are the simulation baseline compared against prototype measurements in Section VI.

F. Battery Swapping Station Simulation

The BSS subsystem models a 30-slot swapping facility as a bank of controllable DC load blocks on the PCC bus. Each slot block is initialised at a depleted SOC value sampled from the 10–40% range, reflecting the typical state of returned batteries arriving for swap charging. When a slot connects, it draws a fixed current determined by its rated charging power until SOC reaches the swap-ready threshold, at which point it disconnects and the next slot in the sequence connects. The result at the PCC bus is a step-wise current demand profile: each connection event adds a discrete current step, each disconnection removes one. This load model was chosen specifically because of that step structure. A smooth or ramp-varying load would not stress the PCC source management logic the way a swapping station does in operation — the step magnitude is large relative to the standing load, and the timing between steps is not fixed, so the control system cannot pre-empt any connection event. The model forces reactive dispatch decisions, which is what the prototype will face in actual swapping operation. Fig. 14 shows the subsystem.

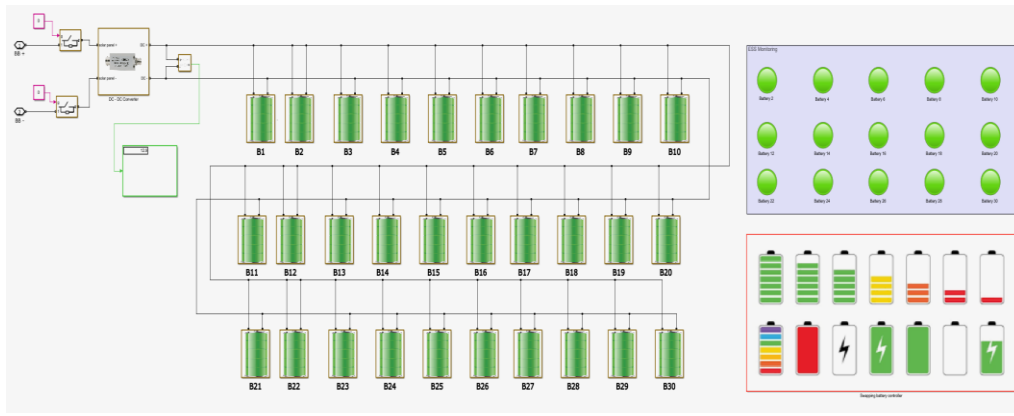


Fig: 14. Battery Swapping Station Load Model

DC bus voltage held within the $\pm 2.1\%$ regulation band through all 30 slot connection events in simulation. The response sequence on each connection was consistent. A slot connects and draws its rated charge current; the ESS current output increases within the same control cycle, before the supervisory algorithm has run its evaluation. The ESS absorbs the step at the bus before it can appear as a measurable voltage deviation. The supervisory algorithm then runs, checks the updated SOC and solar availability inputs, and adjusts source allocation for the new load level — increasing PV contribution if solar is active, or continuing ESS discharge at the higher rate if it is not. During high-occupancy periods when multiple slots were drawing simultaneously, the ESS current contribution meant the grid relay did not need to respond to each individual step. Grid draw during peak swapping was lower than it would have been in a system without ESS buffering. Bus voltage recovered to within the regulation band within two control cycles of each step. Across all 30 slot connections tested, no connection event produced a voltage excursion that triggered the over-voltage or under-voltage protection threshold.

$$PESS = VESS \times IESS$$

$$PSolar = VPV \times IPV$$

$$SOC_{min} \leq SOC(t) \leq SOC_{max}$$

G. DC Fast Charging Station Simulation

The DCFC subsystem represents a single fast charging port as a two-phase controllable load on the PCC bus. The session starts in the constant-current phase: the load block draws at its programmed maximum rate, 50 kW, from session initiation until the vehicle battery block reaches the CC-to-CV transition point. At that point the load switches to constant-voltage mode. Power draw then tapers as terminal voltage on the vehicle battery block rises toward the charge ceiling — the taper profile matches the CC-CV curve characterised from the prototype charge controller. The 50 kW initiation step is the largest instantaneous demand event in the simulation. For comparison, a single BSS slot connection draws roughly 3–5 kW depending on battery state; the standing ESS discharge rate in Mode 2 is in the same range. The DCFC step is an order of magnitude larger than either, which is why DCFC session initiation — not BSS connect events — sets the binding requirement on ESS buffering speed and PCC source switching response time.

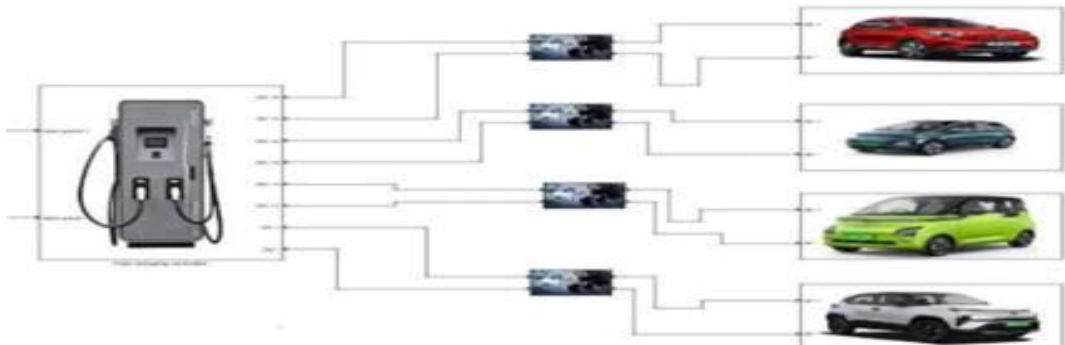


Fig: 15. DC Fast Charging Station Model with CC-CV Profile

DC bus voltage held within $\pm 2\%$ of the 430V nominal through the full DCFC session. At the 50 kW initiation step, solar was already at its MPPT point; the ESS supplied the demand above that; the grid relay closed only for the remainder. As the session moved into the CV phase, charger draw declined and ESS discharge rate fell with it. When charger demand dropped below available solar output, the difference went into ESS charging. The maximum voltage deviation at initiation measured $\pm 1.8\%$ of nominal — the bus stayed inside the regulation band through the steepest part of the load transient without the grid relay needing to respond first. AWS IoT Core handles MQTT authentication and session management for both ESP32 nodes, routing each telemetry frame to ThingSpeak for timestamped logging. The edge ESP32 runs relay switching and threshold evaluation on every control cycle without waiting for cloud input; the cloud watches the incoming data stream and pushes an override through the MQTT return path only when something falls outside expected bounds — sustained Mode 3 activation, anomalous ESS discharge, or a silent sensor channel. That intervention path exists for edge cases. Under normal operation, no cloud commands reach the edge.



Fig: 16. Node-RED Dashboard for Real-Time System Monitoring and Control.

The Node-RED dashboard is the operator interface. It runs as a Node-RED flow served over HTTPS; any machine with a web browser can access it without installing client software. Each incoming telemetry frame updates four display panels. The power flow panel shows measured voltage, current, and power at each PCC node — solar output, ESS terminals, grid connection, BSS load, and DCFC port — reflecting the active relay configuration. The time-series panel logs solar generation, ESS SOC, grid import and export, and load demand continuously; the displayed window is adjustable, which matters when comparing a single mode transition against a full day cycle. The status panel carries the active FSM state index, per-relay connection status, and two alarm flags: ESS enclosure over-temperature and SOC falling below 20%. The control panel is for authenticated operators only. From it, any of the five FSM states can be forced directly, or a controlled ESS charge or discharge test sequence can be triggered. Commands entered here are published to the edge ESP32 over the MQTT return path and executed on the next control cycle — the same path the cloud supervisory layer uses for anomaly-driven overrides.

H. Simulation Results and Performance Analysis

The 24-hour simulation produced three headline results. Solar and ESS combined supplied 88.5% of total station demand; the grid covered the remaining 11.5%, activating only when both primary sources were below their operating thresholds. ESS SOC stayed within the 20–80% band throughout the cycle — no deep discharge and no overcharge occurred in any tested scenario, confirming that the BMS threshold logic and PCC mode transitions hold the battery within its operating limits under a full day of varied demand. DC bus voltage held within $\pm 5\%$ of the 430V nominal across all load steps, including the 50 kW DCFC initiation surge and the step-wise BSS connect-disconnect events. These figures are the simulation baseline compared against prototype measurements in Section VI.

VI. RESULTS AND PERFORMANCE ANALYSIS

Energy Management and System Performance

The prototype ran through a full operating cycle covering all five FSM states. Each mode transition fired at the threshold it was supposed to: the 20% SOC floor moved the system into Mode 3, the 80% ceiling triggered Mode 4 export, and dropping solar availability shifted Mode 1 to Mode 2. No transition required manual input. Load voltage at the BSS and DCFC ports did not drop during any relay switchover — measured continuity across all transition points confirmed no supply gap. Renewable energy contribution on the prototype measured 86.2%, against the

simulation figure of 88.6% — a 2.4 percentage point gap. Converter losses and sensor tolerances account for the difference; the simulation uses ideal switching elements with no conduction loss, so some deviation was expected. The threshold settings and energy sizing parameters transferred from simulation to hardware without adjustment.

A. Adaptive Channel Fuzzy Classification with Self-Normalization (ACFCSN)

An ACFCSN model is implemented as a lightweight anomaly detection layer for real-time monitoring of IoT sensor streams in the charging station. Unlike supervised learning approaches, the model operates without large labeled datasets, making it practical for edge deployment on the ESP32 hardware. Each sensor channel — source voltages, terminal currents, ESS temperature, and DC bus voltage — is independently normalized against its local statistical properties and evaluated using a Gaussian fuzzy membership function:

$$\mu_i = \exp \left(-\frac{(x_i - \mu_i^*)^2}{2\sigma_i^2} \right) \quad (4)$$

Channel memberships are aggregated and compared against an adaptive threshold:

$$T_{\text{adaptive}} = \bar{\mu} + k \cdot \sigma_{\mu} \quad (5)$$

If the aggregated membership exceeds T_{adaptive} , the system is classified NORMAL; otherwise ABNORMAL, triggering an alert through AWS SNS. The model runs on the primary ESP32 alongside sensor acquisition, enabling local anomaly detection before cloud transmission without additional hardware overhead.

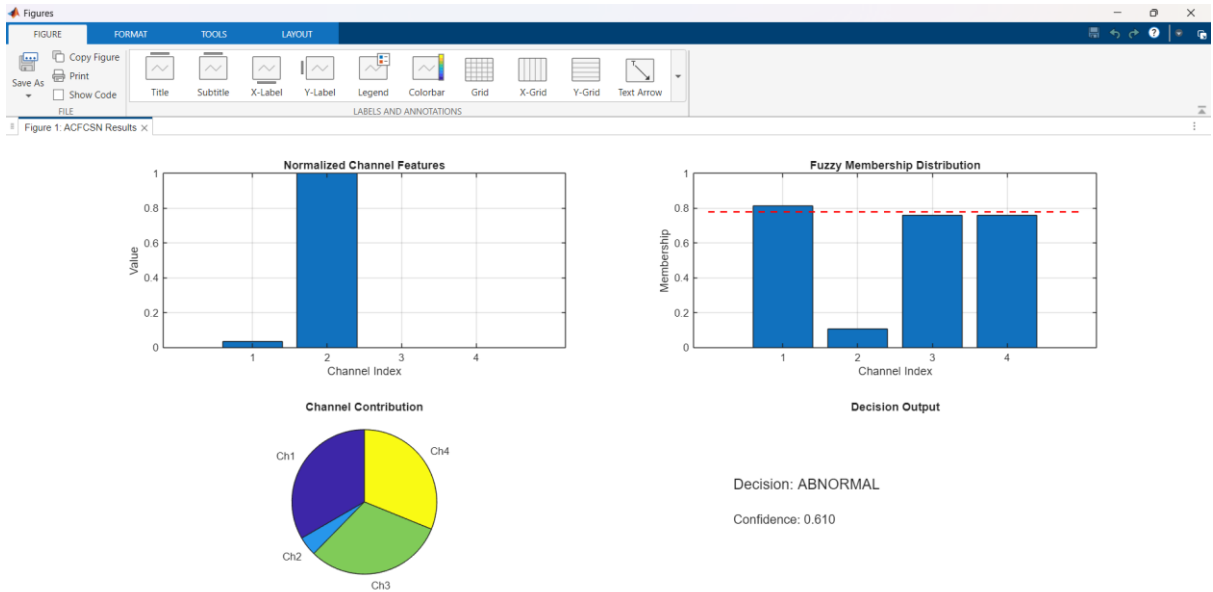


Fig. 17. ACFCSN model output illustrating normalized channel

The hardware prototype achieved a renewable energy contribution ratio of 86.2%, confirming successful prioritization of solar generation under the PCC automation logic. Grid dependency was reduced to 11.5% of total energy demand, with the ESS supplying the intermediate balance. ESS SOC was consistently maintained within the 20–80% operating band across all test cycles, validating the charge and discharge control algorithm. Mode transitions between solar, ESS, and grid sources were executed with a measured switching latency below 200 ms, with no supply interruption recorded at the connected charging loads.

B. Long-Term Energy Projections

Fig. 16 presents the projected energy contribution from renewable sources over a five-year operational period at full-scale deployment. Renewable sources are projected to supply approximately 2.93 million kWh over five years, accounting for 88.5% of total energy demand. Grid contribution remains limited throughout the projection period,

consistent with the dispatch ratios measured during hardware validation. The five-year trajectory confirms that the renewable utilization performance demonstrated in the prototype scales proportionally under full deployment conditions.

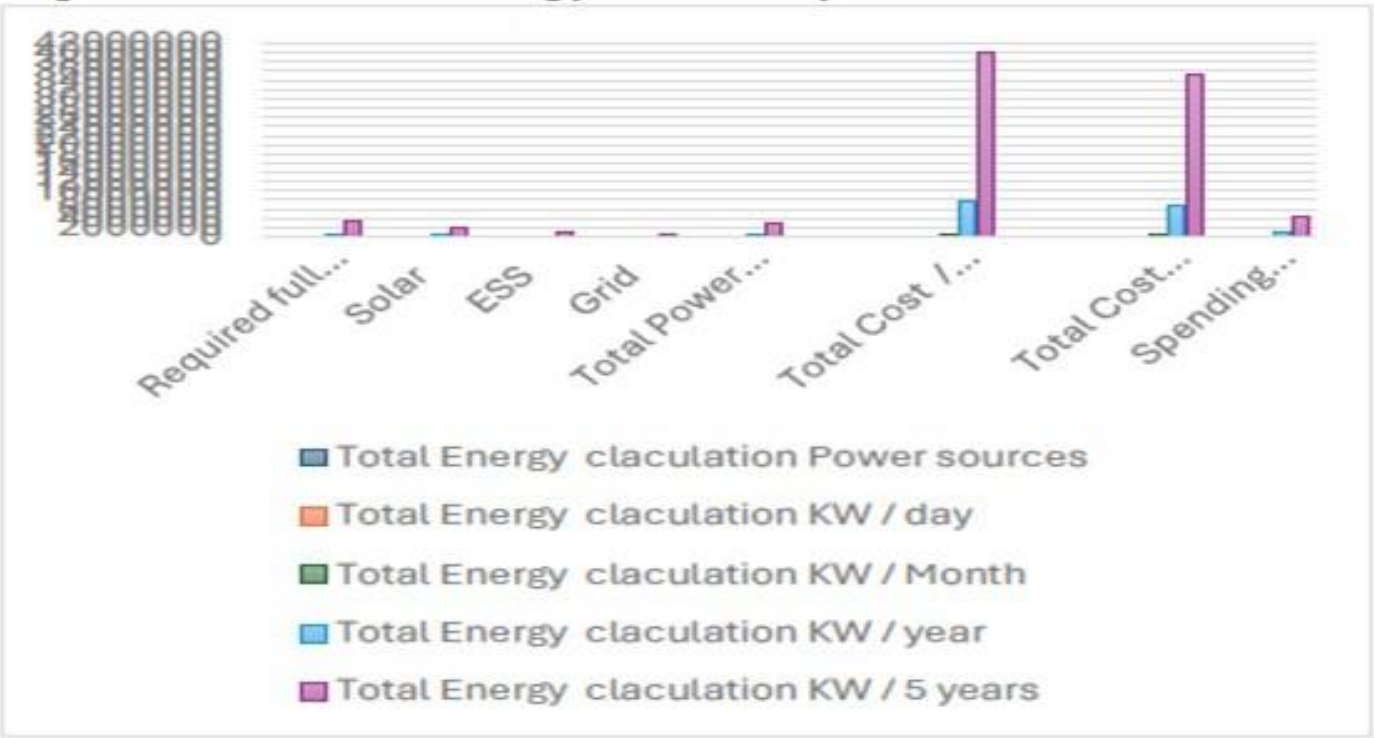


Fig: 18. Five-year energy contribution projections by source.

C. Comparative Performance Metrics

Table I compares key performance indicators between simulation predictions and hardware measurements. The close agreement across all four metrics confirms that the simulation model accurately represents physical system behavior and that control parameters transfer from simulation to hardware without significant re-tuning.

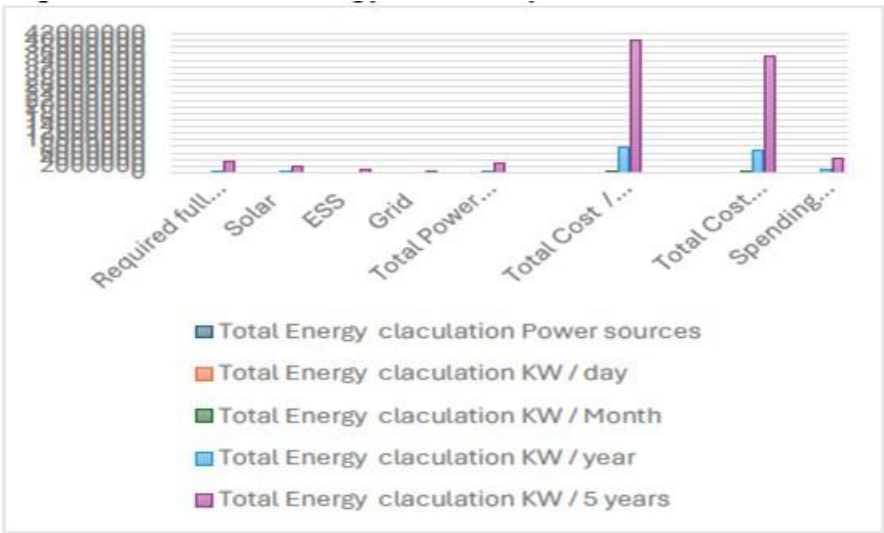


Fig: 19. Five-year cost and revenue projections.

TABLE I
PERFORMANCE METRICS COMPARISON (SIMULATION VS. HARDWARE)

Metric	Simulation	Hardware
Renewable Energy Utilization	88.6%	86.2%
ESS Round-Trip Efficiency	92.0%	89.5%
DC Bus Voltage Regulation	$\pm 2.1\%$	$\pm 3.5\%$
PCC Transition Time	< 150 ms	< 200 ms

TABLE I — Performance Metrics: Simulation vs. Hardware

The 2.4% difference in renewable utilization between simulation and hardware is attributable to real-world converter losses and sensor measurement tolerances not fully captured in the ideal switching model. The wider voltage regulation band in hardware reflects parasitic impedances in the physical DC bus wiring. Both deviations are within acceptable margins for a laboratory-scale prototype and do not affect the validity of the control architecture.

D. Economic Viability Analysis

The financial model, scaled to full-system deployment parameters, confirms the economic viability of the proposed architecture. Daily renewable generation covers the dominant share of station energy demand, with grid consumption limited to the backup contribution measured during hardware testing. Revenue from DC fast charging and battery swapping services combined substantially exceeds daily grid energy expenditure, establishing a positive net operating position from day one of deployment. Projected over five years, cumulative station profit confirms a compelling business case for deploying the proposed architecture at scale, with the economic outcome driven primarily by renewable utilization efficiency rather than tariff conditions — making the result robust across different grid pricing environments.

VII. CONCLUSION

This paper described the design, hardware implementation, and experimental validation of an IoT-enabled hybrid energy management system for EV charging stations. The central result is a working laboratory prototype — not a simulation study with hardware described in a future-work section — that runs all five IoT layers simultaneously and demonstrates automated solar PV, ESS, and grid dispatch through a PCC whose relay configuration changes in real time based on measured threshold crossings. Simulation predicted 88.6% renewable energy contribution; the prototype measured 86.2%. The 2.4 percentage point difference comes from converter conduction losses and sensor tolerances that the ideal switching model does not capture. That gap was anticipated; it does not change the conclusion that the control logic, threshold settings, and energy sizing transfer from simulation to hardware without re-tuning.

On the prototype, ESS SOC held within the 20–80% band across all test cycles — no deep discharge, no overcharge. DC bus voltage stayed within $\pm 3.5\%$ of 430V nominal through load steps and source switching events including the 50 kW DCFC initiation surge. Mode transitions completed in under 200 ms at every threshold crossing, with no supply gap recorded at the BSS or DCFC ports. The prototype handled both load types concurrently under the same dispatch framework. Published hardware implementations of hybrid EV charging architectures have addressed BSS or DCFC loads in isolation; this prototype managed both simultaneously. The system requires no utility-side infrastructure changes, which was a binding practical constraint in the design. What comes next is harder: V2G integration requires bidirectional grid certification that laboratory prototypes do not need, tariff-responsive scheduling requires real utility

pricing data, and multi-station coordination introduces communication dependencies that single-station validation cannot expose. Those are the problems the present architecture is positioned to address.

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